

Shenzhen Polcd Technology Co.,Limited	Rev No	Issued Date.	Page
	A	2024.07.08	1/13

Project Size.	1.54 inch		
Model No.	P154H007-12		
Samples No.			
Product type.	240xRGBx240 SPI mode		
Signature by customer:			
Prepared	Checked	Approved	

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Shenzhen Polcd Technology Co.,Limited	Rev No	Issued Date.	Page
	A	2024.07.08	2/13

1.0 GENERAL DESCRIPTION

Item	Specification	Unit
Screen Size	1.54 inch	Diagonal
Number of Pixel	240RGB(H)x240(V)	Pixels
Display area	27.72(H)x27.72(V)	mm
Pixel pitch	0.115(H)x0.115(V)	mm
Outline Dimension	31.52x33.72x1.86	mm
Pixel arrangement	RGB Vertical Stripe	— —
Display mode	Normally Black	— —
Viewing Direction(eye)	ALL	— —
Gray inversion direction	--	
Display Color	262K	— —
Luminance(cd/m ²)	400	nit
Contrast Ratio	1100:1	— —
Surface treatment	— —	— —
Interface	4W SPI	
Back-light	LED Side-light type	— —
LCM Drive IC	ST7789P3	
CTP Drive IC		
Operation Temperature	-20~70	°C
Storage Temperature	-30~80	°C
Weight	— —	g

1.1 Features

- 4W SPI interface.

1.2 Applications

- MPOS Device.
- Personal Navigation Device.
- Other devices which require high quality displays.

2.0 INPUT INTERFACE PIN ASSIGNMENT

FPC connector is used for electronics interface.

PinNo.	Symbol	Function
1	GND	Ground
2	GND	Ground
3	FMARK	
4	SDA	SPI interface input/output pin
5	SCL	Serial interface clock
6	RS	Display data/command selection pin in parallel
7	CS	Chip select input pin
8	RESET	External reset input.
9	LEDA	LED back light(Anode)
10	LEDK	LED back light(Cathode)
11	IOVCC	Power Supply
12	VCC	Power Supply

3.0 OPTICAL CHARACTERISTICS

3.1 Optical specification

Item		Symbol	Condition	Min	Type	Max	Unit	Note
White luminance (Center)		Lv	Θ=0 Normal Viewing Angle I _{BL} =40mA	--	400	--	cd/m ²	(4)(5)(7)
Response time		Tr+Tf		--	30	35	ms	(3)
Contrast ratio		CR		900	1100	--	--	(2)(4)
Color Chromaticity (CIE1931)	white	Wx		0.304	0.324	0.344		(6)
		Wy	0.327	0.347	0.367			
Viewing Angle	Hor	ΘL	CR≥10	80	85	--		(1)
		ΘR		80	85	--		
	Ver	ΘU		80	85	--		
		ΘD		80	85	--		
Brightness uniformity		Avg	Θ=0	80	90	--	%	(5)
Color Gamut		NTSC	Θ=0	55	60	--	%	(6)
Optima View Direction		ALL						(1)

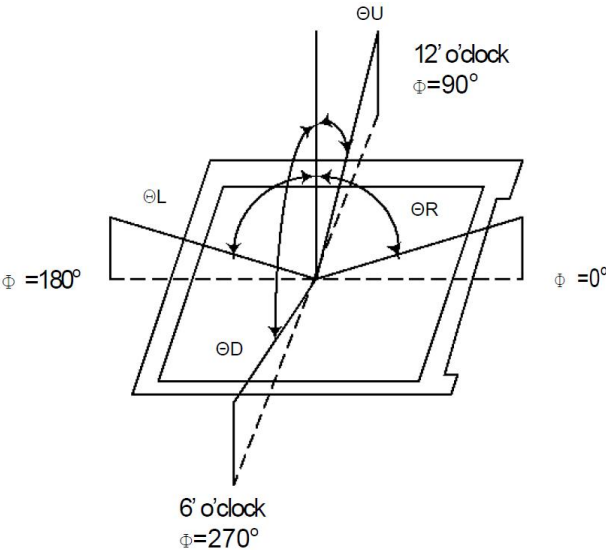
3.2 Measuring Condition

- Measuring surrounding: dark room
- LED current IL: 40mA
- Ambient temperature: 25±2℃
- 15min. warm-up time

3.3 Measuring Equipment

- FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-7 for other optical characteristics.
- Measuring spot size: 20 ~ 21 mm

Note (1) Definition of Viewing Angle

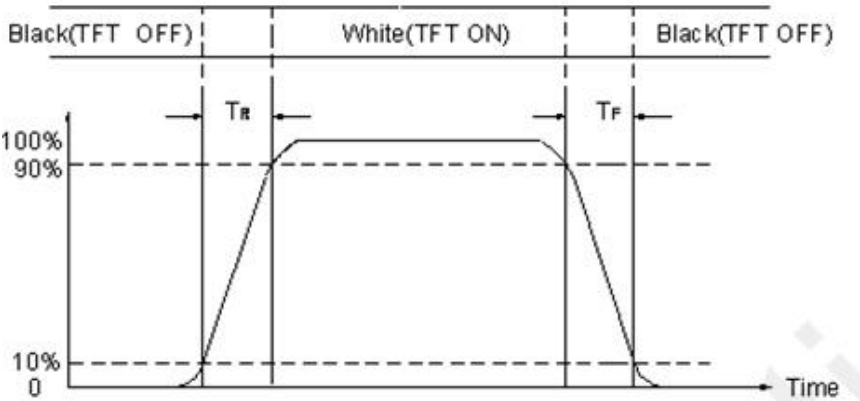


Note (2) Definition of Contrast Ratio(CR):

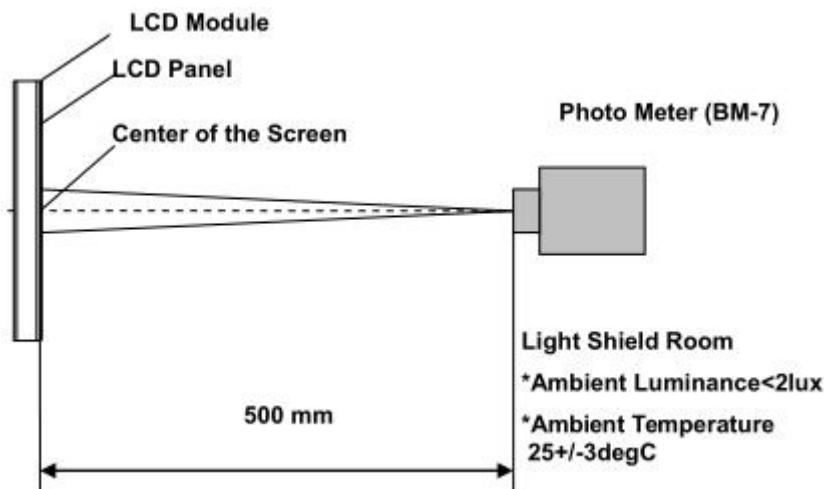
Measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

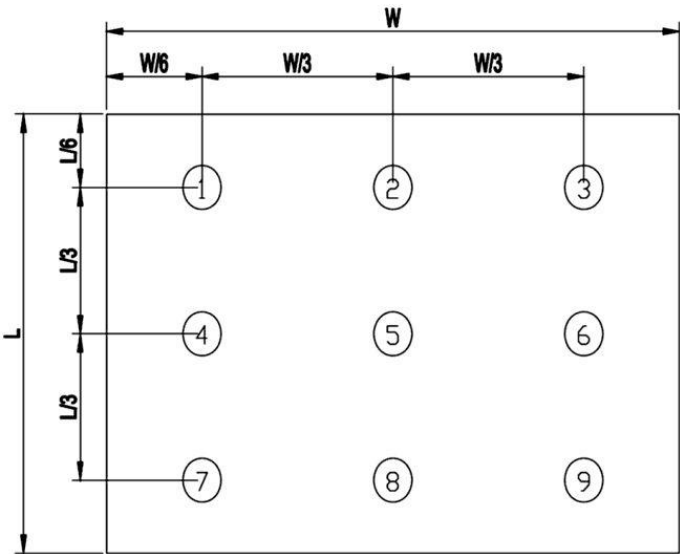
Note (3) Definition of Response Time: Sum of TR and TF



Note (4) Definition of optical measurement setup



Note (5) Definition of brightness uniformity
The luminance uniformity is calculated by using following formula.
 $\Delta Bp = Bp \text{ (Min.)} / Bp \text{ (Max.)} \times 100 \text{ (\%)}$
Bp (Max.) = Maximum brightness in 9 measured spots
Bp (Min.) = Minimum brightness in 9 measured spots .



Note (6) Definition of Color of CIE1931 Coordinate and NTSC Ratio.
Color gamut:

$$S = \frac{\text{Area of RGB triangle}}{\text{Area of NTSC triangle}} \times 100\%$$

4.0 ELECTRICAL CHARACTERISTICS

4.1 TFT LCD Module

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Analog supply voltage	VDD	2.4	2.8	3.3	V	
Digital supply voltage	VDDI	1.65	1.8	3.3		
Input signal Voltage	VIH	0.7VDDI	-	VDDI	V	
	VIL	GND	-	0.3VDDI	V	

4.2 Back-Light Unit

The backlight system is an edge-lighting type with 2 LED Dies.
The characteristics of the LED are shown in the following tables.

Item	Symbol	Min	Typ	Max	Unit	Note
LED current	IL	-	30	40	mA	(2)
LED voltage	VL	-	2.8	3.2	V	
Operating LED life time	Hr	-	20000	15000	Hour	(1)(2)

Note (1) LED life time (Hr) can be defined as the time in which it continues to operate under the condition: Ta=25±3 °C, typical IL value indicated in the above table until the brightness becomes less than 50%.

Note (2) The “LED life time” is defined as the module brightness decrease to 50% original brightness at Ta=25°C and IL=20mA.

5.0 SPI Interface Timing Characteristics

7.4.3 Serial Interface Characteristics (4-line serial):

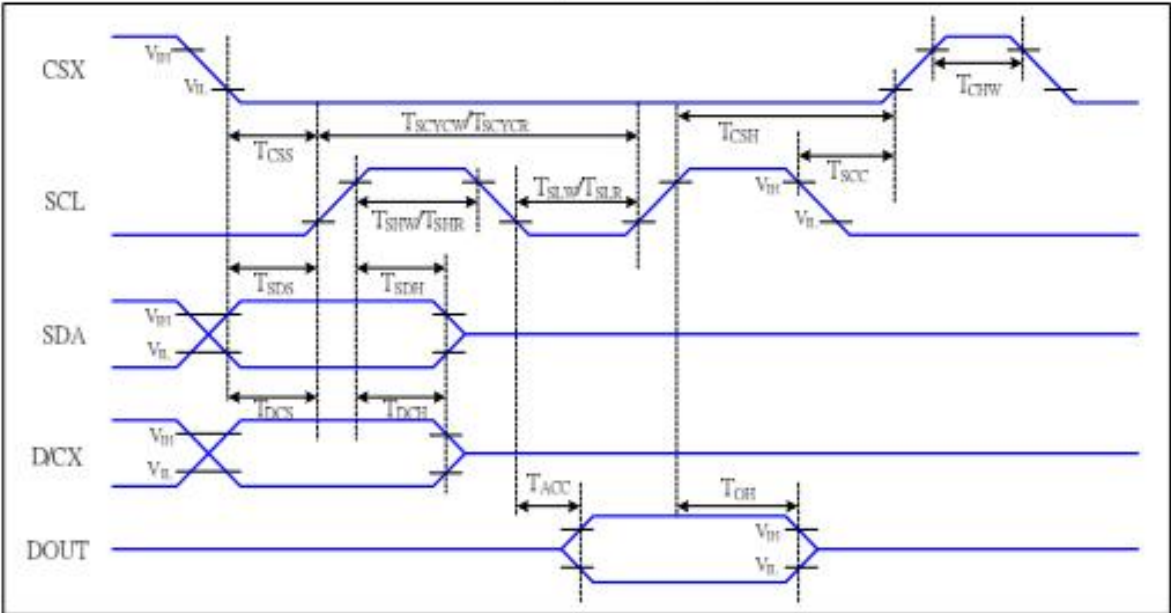


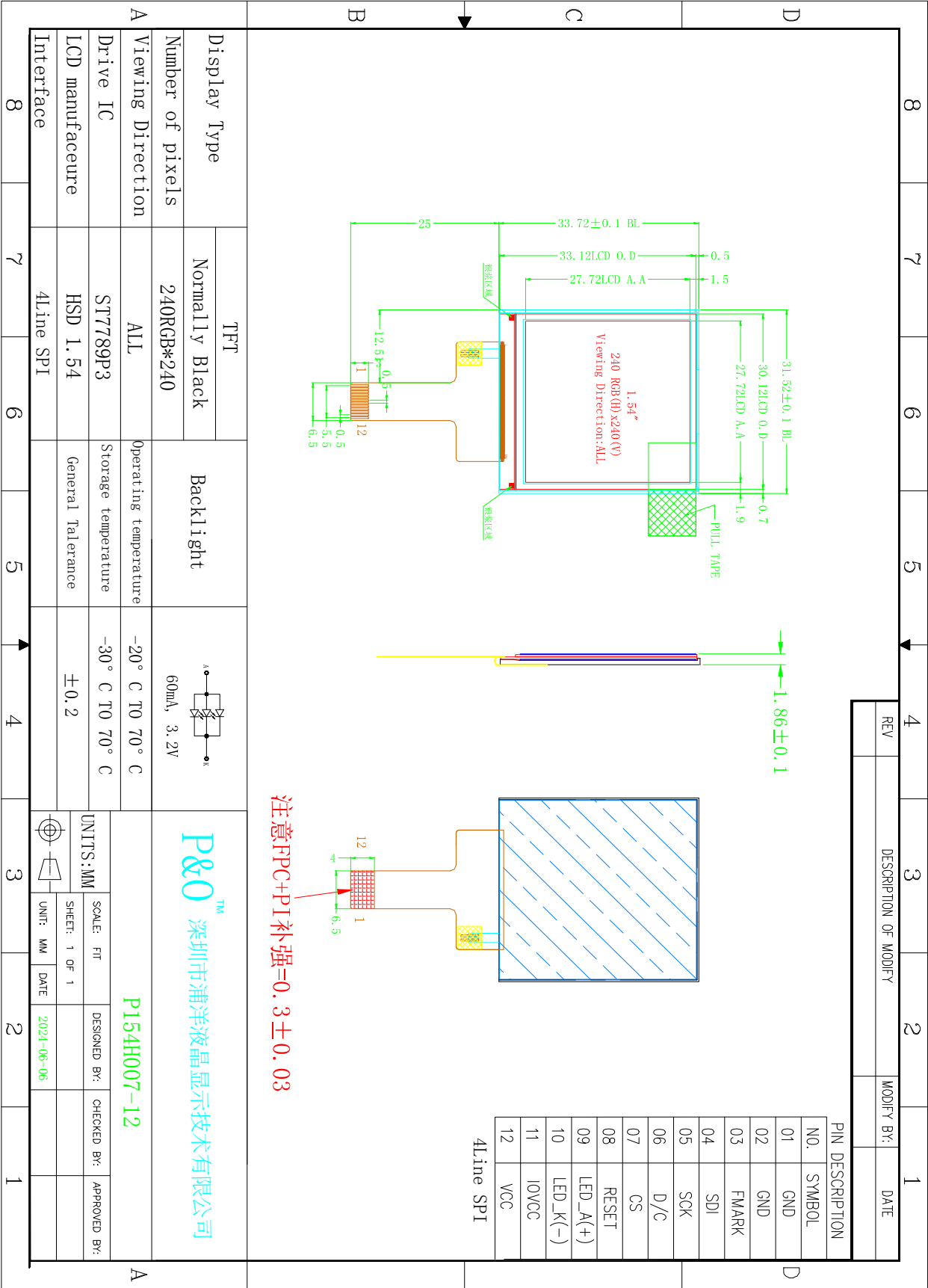
Figure 5 4-line serial Interface Timing Characteristics

VDD1=1.65 to 3.3V, VDD=2.4 to 3.3V, AGND=DGND=0V, Ta=-30 to 70 °C

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SDC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select "H" pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	66		ns	-write command & data ram
	T_{SHW}	SCL "H" pulse width (Write)	15		ns	
	T_{SLW}	SCL "L" pulse width (Write)	15		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	-read command & data ram
	T_{SHR}	SCL "H" pulse width (Read)	60		ns	
	T_{SLR}	SCL "L" pulse width (Read)	60		ns	
D/CX	T_{DCS}	D/CX setup time	10		ns	
	T_{DCH}	D/CX hold time	10		ns	
SDA (DIN)	T_{SDS}	Data setup time	10		ns	
	T_{SDH}	Data hold time	10		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Table 6 4-line serial Interface Characteristics

6.0 OUTLINE DIMENSION



Shenzhen Polcd Technology Co.,Limited	Rev No	Issued Date.	Page
	A	2024.07.08	10/13

7.0 Items and Criteria:

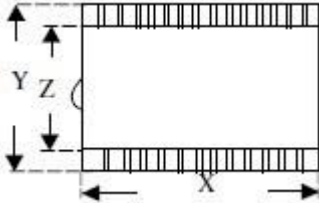
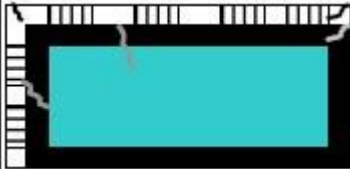
7.1 Guarantee

APEX warrants the quality of our products for **1 year** (from the date of delivery). If there are functional defects found during the period of warranty, the defective products would be replaced on a one-to-one basis. Apex would not be responsible for any direct /indirect liabilities consequential to any parties.

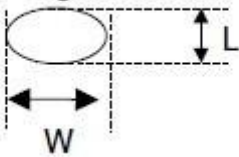
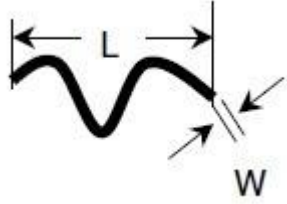
All the products should be stored or used as specified conditions described in these sheets. If module productions are not stored or used as specified conditions, herein, it will be void the **1 year** warranty(guarantee).

7.2 Visual inspection criterion in cosmetic

(1) Glass defect

Glass defect			
NO	Defect	Criteria	Remark
1	Dimension(Minor)	By engineering diagram	
2	Cracks(Major)	Extensive crack 【Reject】	

(2) LCM appearance defect

NO	Defect	Criteria		Remark
1	Round type(Minor)	Spec	Permissible Qty	1. $\psi = (L+W)/2$, L: Length, W: Width 2. Disregard if out of A.A. 
		$\psi \leq 0.10\text{mm}$	Disregard	
		$0.10\text{mm} < \psi \leq 0.20\text{mm}$	3	
		$0.20\text{mm} < \psi$	0	
2	Line type(Minor)	Spec	Permissible Qty	1. L: Length, W: Width 2. Disregard if out of A.A. 
		$W \leq 0.03\text{mm}$	Disregard	
		$L \leq 3.0\text{mm}$ and $0.03\text{mm} < W \leq 0.05\text{mm}$	2	
		$L \leq 3.0\text{mm}$ and $0.05\text{mm} < W \leq 0.10\text{mm}$	1	
		$W > 0.10\text{mm}$ or $L > 3.0\text{mm}$	0	
3		Spec.	Permissible Qty	1. $\psi = (L+W)/2$, L: Length, W: Width 2. Disregard if out of A.A.
		$\psi \leq 0.20\text{mm}$	Disregard	

	Polarizer	0.20mm<ψ ≦ 0.30mm	2	
	dent(Minor)	0.30mm<ψ ≦ 0.50mm	1	

(3) FPC

NO	Defect	Criteria	Remark
1	Copper peeling(Minor)	Copper peeling 【Reject】	
2	Golden finger	FPC golden finger broken, dead fold, indentation makes FPC surface broken 【Reject】 Tin plating layer(or gold plating) scratch, but not hurt circuit 【Accept】 Except circuit, other position scratch but not expose metal wire 【Accept】	
3	Pin	FPC PI layer delamination 【Reject】 Material and color are inconsistent with sample, FPC burrs 【Reject】 FPC Pin deformation but not affect function. 【Accept】 FPC Pin area is dirty 【Reject】 Other than FPC Pin area is dirty but not affect function 【Accept】	
4	Golden finger	Golden finger edge has burrs,foreign material 【Reject】 Golden finger oxidation (dark), uneven electroplating, pinhole, foreign material 【Reject】 Golden finger soldering pad crack exceeds 1/3 length of soldering pad, and soldering pad crack exceed 2 Pins 【Reject】 Golden finger tin plating(or gold plating)scratch, but not hurt circuit 【Accept】 Other than golden finger area scratch but not expose metal circuit 【Accept】	
5	FPC Silk printing	Ghosting, incomplete silk printing, wrong printing 【Reject】	
6	FPC Circuit line width	Line width deviation exceed 1/3 line width 【Reject】	

(4) Black tape

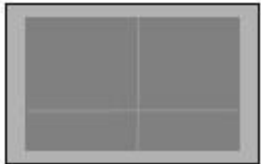
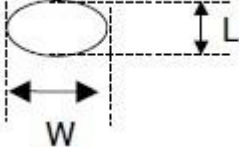
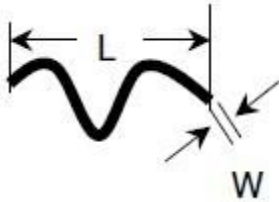
NO	Defect	Criteria	Remark
1	Shift(Minor)	IC exposed 【Reject】	
2	No black tape(Minor)	No black tape 【Reject】	

(5) Silicon

NO	Defect	Criteria	Remark
1	Amount of silicon (Minor)	ITO exposed 【Reject】	

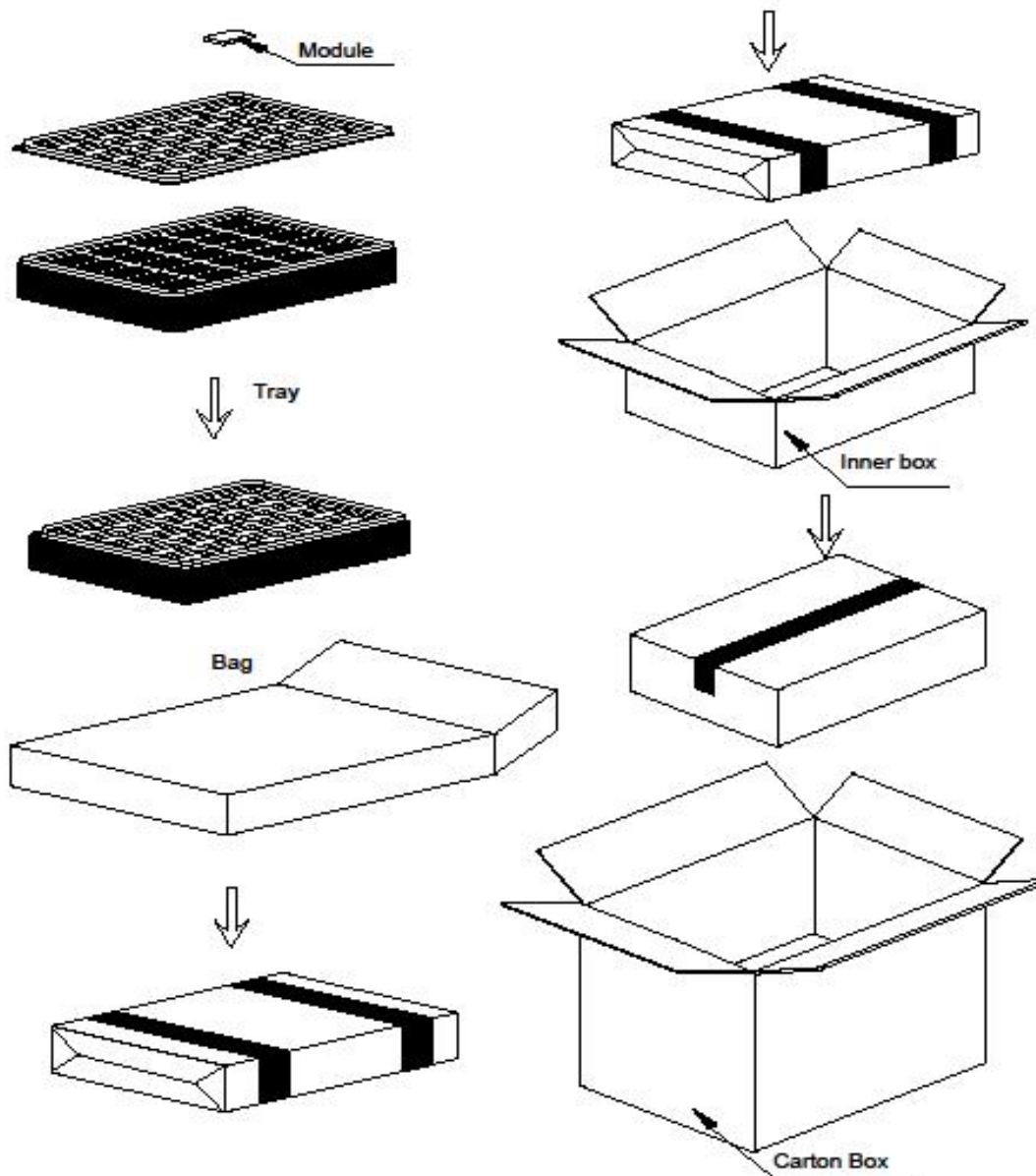
Shenzhen Polcd Technology Co.,Limited	Rev No	Issued Date.	Page
	A	2024.07.08	12/13

7.3 Visual inspection criterion in electrical display

NO	Defect	Criteria		Remark
1	No display (Major)	Not allowed		
2	Missing line (Major)	Not allowed		
3	Darker or lighter Line (Major)	Not allowed		
4	Weak line(Major)	By limited sample		
5	Bright / Dark point (Minor)	Spec.	Permissible Qty	1:1sub-pixel: 1R or 1G or1B 2:Point defect area $\geq 1/2$ sub pixel.
		Bright point	1	
		Dark point	2	
6	Round type (Minor)	Spec	Permissible Qty	1. $\psi=(L+W)/2$, L: Length, W: Width 2. Disregard if out of A.A. 
		$\psi \leq 0.10\text{mm}$	Disregard	
		$0.10\text{mm} < \psi \leq 0.20\text{mm}$	3	
		$0.20\text{mm} < \psi$	0	
7	Line type (Minor)	Spec.	Permissible Qty	1. L: Length, W: Width 2. Disregard if out of A.A. 
		$W \leq 0.03\text{mm}$	Disregard	
		$L \leq 3.0\text{mm}$ and $0.03\text{mm} < W \leq 0.05\text{mm}$	2	
		$L \leq 3.0\text{mm}$ and $0.05\text{mm} < W \leq 0.10\text{mm}$	1	
		$W > 0.10\text{mm}$ or $L > 3.0\text{mm}$	0	
8	Mura (Minor)	By 5% ND filter invisible		

8. Packing (Reference only)

Packing Method



1. Put modules into tray
2. Stack the trays
3. Pack the tray stack with a bag and fix with adhesive tape
4. Put the tray stack into inner box and seal with adhesive tape.
5. Put inner boxes into carton and seal with adhesive tape.